

Zixun Huang

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RESEARCH INTEREST

Theoretical Foundations of Machine Learning, Scaling Laws, Reinforcement Learning Theory

EDUCATION

- **Peking University, School of Mathematical Sciences** Sept. 2022 – Present
 - *B.S. in Statistics*, GPA: 3.68/4.0
 - Member of Elite Undergraduate Program for Applied Mathematics (**Top 10%**)
 - **Core Coursework:** Mathematical Analysis (94), Advanced Algebra (97.5), Mathematical Modeling (91), Numerical Methods (88), Deep Learning Theory (87), Learning Data Science with Python (90), Mathematical Statistics (97)
- **University of California, Berkeley** Jan. 2025 – Aug. 2025
 - *Visiting Student*, GPA: 4.0/4.0
 - **Core Coursework:** Theoretical Statistics (A+), Advanced Topics in Probability and Stochastic Processes (A), Modern Statistical Prediction and Machine Learning (A+), Structure and Interpretation of Computer Programs (A+)

PUBLICATIONS

- Binghui Li*, Fengling Chen*, **Zixun Huang***, Lean Wang*, Lei Wu. *Unveiling the Role of Learning Rate Schedules via Functional Scaling Laws*. **NeurIPS 2025 (Spotlight)**. [\[Paper\]](#)
- **Zixun Huang***, Jiayi Sheng*, Zeyu Zheng. *OBLR-PO: A Theoretical Framework for Stable Reinforcement Learning*. Submitted to **AISTATS 2026 (Reviewer scores: 6, 5, 5 (max 7))**.

RESEARCH EXPERIENCE

- **Functional Scaling Laws for Learning Rate Schedules (Core)** May. 2024 – Nov. 2025
 - *Advisor:* [Prof. Lei Wu](#)
 - Proposed a novel Functional Scaling Law (FSL) by modeling SGD as an intrinsic-time SDE
 - Analyzed learning-rate schedules under data and compute constraints, revealing efficiency trade-offs
 - Validated FSL through kernel regression and LLM pre-training, producing optimal schedules
- **OBLR-PO: Theory for Stable Reinforcement Learning (Core)** Feb. 2025 – Oct. 2025
 - *Advisor:* [Prof. Zeyu Zheng](#)
 - Established theoretical guarantees for policy optimization, including loss bounds and convergence
 - Introduced adaptive learning rate and gradient-weighted baselines for principled variance reduction
 - Implemented OBLR-PO and demonstrated superior performance on Qwen3-4B/8B vs. GRPO
- **Combining Simulation with Linear Algebra: COSIMLA** May. 2025 – Jul. 2025
 - *Advisor:* [Prof. Zeyu Zheng](#)
 - Implemented and evaluated five COSIMLA estimators across diverse parameter regimes
 - Analyzed variance-bias-efficiency trade-offs, confirming improved stability and speed

- **Saturation Phenomenon Under Kernel Method**

Feb. 2024 – May. 2024

– *Advisor: Prof. Lei Wu*

– Derived loss functions for kernel ridge regression and gradient descent; analyzed the saturation phenomenon through bias–variance tradeoff and Fourier perspective

– Identified a noiseless setup saturation effect previously overlooked in prior work

• **Cross Modal Alignment**

May. 2024 – Jun. 2024

– *Advisor: Prof. Junfeng Hu*

– Developed an alignment strategy for movie posters and text introductions using vector embeddings

– Established baseline accuracy and reported insights on model and network pre-training effects

• **Analysis for Grokking Phenomenon**

Nov. 2023 – Jan. 2024

– *Advisor: Prof. Lei Wu*

– Replicated grokking phenomenon experiments (MLP, LSTM) on modular addition task

– Conducted theoretical and empirical analysis from training-dynamics and landscape perspectives
- SELECTED HONORS AND AWARDS

- Hong Sheng Scholarship (**Top 12%**)

Sept. 2024

• Yau Contest Groups Prize (**Top 5%**)

May. 2024

• First Prize, Chinese Mathematics Contest (**Top 1%**)

Apr. 2024

• Xiaomi Scholarship (**Top 12%**)

Sept. 2023

• Gold Prize, Chinese Mathematical Olympiad

Dec. 2021

• Silver Prize, Chinese Mathematical Olympiad

Nov. 2020
- EXTRACURRICULAR ACTIVITIES

- **Applied Mathematics Seminar, Peking University**

Jun. 2024 – Jul. 2024

– Teaching Assistant for *Prof. Zhi-Qin John Xu*

– Assisted in developing and organizing lecture materials

• **Peking University Runners Association**

Oct. 2022 – Dec. 2023

– Core Team Member

– Organized and coordinated running events; participated in volunteer activities

– Completed the Nanjing Half Marathon
- SKILLS AND OTHERS

- **Languages**

Chinese (Native), English (Fluent)

• **TOEFL iBT**

Total 101: 28(R), 23(L), 23(S), 27(W)

• **Quantitative Skills**

GRE General: V160, Q170; GRE Math: 970 (95th percentile)

• **Programming**

Python, MATLAB, LaTeX, PyTorch, C++, Go, R, SQL